

A05 Written Report

Analyzing *Arrival* Through the Lens of Natural Language Processing

Movie: Arrival (2016)

Team Collaboration Draft

Team members: Jonah, Fifehanmi, Justin, Brandon, Samantha, and Elijah

Team Writing Sections

Jonah — Movie Overview

Your section:

In the movie *Arrival* (2016), Louise Banks is a linguist brought in by the US military to try and communicate with aliens from space named Heptapods. The Heptapods speak a different language from humans, so her job is to decipher their language and what they are trying to accomplish on Earth. One specific scene where communication is shown is one of the first times Louise has tried to communicate with the Heptapods. She writes the word “Human” on a board and points to herself and her partner. She then tells them her name and requests that they respond in their own language. The Heptapods then spray a symbol on the glass as a response, and Louise begins to try and decipher what the symbol means. This scene shows how Louise wants to build a basic understanding of English with the Heptapods through simple words and repetition.

This movie connects to NLP because computers also try to look for patterns and context to understand language. In the same way Louise was trying to decipher the Heptapods symbols; an NLP system can dissect a sentence by analyzing the surrounding words to get the full context of the sentence. There is another scene later in the movie in which one of the Heptapods messages is translated to “weapon”, which throws the whole world into a panic. It is later found that the translation was more akin to “tool” in which they were offering mankind. So, this shows how simply translating is not the key to communication, rather it is more important to understand the intention behind the words and what they mean.

Sources used: *Arrival* (2016), directed by Denis Villeneuve, Paramount Pictures

Fifehanmi — NLP Challenges

Your section:

Honestly, the scene that stuck with me most is when the team finally translates one of the heptapods' messages as "offer weapon," and almost everyone jumps straight to the worst possible reading. China and a few other countries take it as a threat and start preparing for conflict, while Louise keeps pushing back since, in their language, the symbol they translated as "weapon" could just as easily mean "tool" or "gift." What makes the message so hard to understand isn't really the symbols themselves; it's more

that nobody has enough context concerning how the heptapods use language to know which meaning they intended, so the same two words could mean "use this weapon against each other," "we're armed," or "we're giving you something useful." Each reading led somewhere different, whether that was the cooperation between the twelve countries hosting the shells, the panic among soldiers and the public, or the war between humans and the heptapods, which is pretty much what almost happened. In this case, the weapon was the language itself. Which makes sense when you hear it, but nobody thought about it until it was almost too late.

This ties directly into word-sense disambiguation and translation in NLP, since a system that picks the most common meaning of a word without considering context or the speaker's intent can change the whole message. A translation model that sees "weapon" and defaults to its usual meaning would make the same mistake the military did, and a sentiment model would probably flag the message as hostile even though it wasn't meant that way. That said, I don't think this is only an AI problem, since the people in the movie fail for the same reason a model would: they fill in the lacking information with their own fears and cultural assumptions instead of figuring out what the speaker actually meant. It reminds me of getting a short text from a distant friend: you understand every word, yet you can't tell whether they're being ironic or genuine, so you end up guessing based on your mood.

Hence, for both people and AI, understanding language is less about knowing what the words mean and more about knowing enough about the speaker, the situation, and the culture behind it to choose the right meaning, which is exactly what Louise understood and what everyone else was too quick to skip past.

Sources used:

Arrival (2016), directed by Denis Villeneuve, Paramount Pictures;

Justin — How the Alien Language Is Learned

Your section:

In the film *Arrival* (2016), Dr. Louise Banks and her team tasks themselves with deciphering the language of the aliens using methods that are commonly used in the AI Natural Learning process.

1. Simple Words Atomic Tokens & Morphemes

Louise focuses on establishing basic words and concepts when she presents a whiteboard with only basic words combined with the person, object, or action; it establishes the core blocks of communication. This is the exact method used for tokenization to build a vocabulary for NLPs. NLPs text data is broken down into tokens, characters and sub-words for the model to understand.

2. Visuals & Grounding Multimodal Data & Embedding Alignment

To teach the aliens what "Louise" or "Human" means, she first points to herself then writes "Louise" on a screen. Using this method, she can pair the word with a meaning (Louise = that person). This method is similar to the Symbol Grounding Problem combined with Multimodal Machine Learning. A text-only model views symbols without knowing what they mean and the Multimodal used the symbols viewed as tokens and combines it with a visual/physical feature to understand the meaning of the word

3. Repetition Large-Scale Dataset Training & Frequency Analysis

The decipherment process is not instant; it takes an extremely long time and takes time to test and retest the same written prompts to record the results of each attempt to finally decipher the language. Just like the deciphering process in Arrival, NLP systems take time and repeat examples for it to learn. This process is often described as massive data repetition (or training).

4. Patterns & Geometry Syntax, Feature Extraction & Self-Attention

Ian Donnelly analyzes the Heptapod logo, He measures how thick strokes are, the total ink density, the curvature of symbols, and common geometry. This effort results in Ian discovering that the logo combines multiple thoughts without any beginning or end to it. The way Ian analyzes the logo mirrors how feature extraction and self-attention Mechanics work. These systems use attention mechanisms to understand the relationship between all words used.

5. Contextual Embeddings & Latent Space

Over time, Louise is able to craft a dictionary by combining simple words, repetition visual representation, and pattern recognition together. Overtime, these symbols transform from simple words or ideas into complex ones, which allows the team to decode complex statements which is similar to how Models go from static representation to vector embedding. NLPs assign a static vector to a word while deep NLPs build meaning with deep neural layers.

Sources used: [Arrival via YouTube]

Brandon — NLP Approaches

Your section:

Scene Analysis

- For my section, I analyzed the scene where Louise and Ian step inside the vessel to establish contact, and the heptapods shoot a burst of ink against the glass wall to form

their very first circular logogram. This single moment shows how hard and complex alien communication is, but it also gives us a clear look at how different NLP methods would tackle the language.

Rule-Based NLP

- **Movie Example:** Think about the scene where Louise brings out the whiteboard and starts teaching the aliens basic words like "*Louise*," "*human*," or "*walk*."
- **The Connection:** This is basically a manual version of **rule-based NLP** and **tokenization**. Before a computer can translate anything, it has to break a sentence down into single pieces or "tokens" (like individual words). Louise is doing exactly that by hand. She's creating a basic rulebook from scratch so both sides can understand what a specific word means without any confusion.

Statistical NLP

- **Movie Example:** Later on, the team sets up those computer screens in the tents to scan all the different circular symbols the aliens make, tracking how the shapes change.
- **The Connection:** This is a lot like **statistical NLP**. Instead of following strict grammar rules, statistical systems look at tons of data to find patterns. The software in the movie is calculating how often certain squiggles repeat or show up together. By looking at that frequency, the computer helps them guess the most likely meaning of a symbol based on past data.

Deep NLP

- **Movie Example:** The aliens don't write in a straight line from left to right. They blast out a full circular symbol all at once, packing a whole sentence into one image. The team has to use **computer vision** to capture these visual symbols in real-time.
- **The Connection:** This maps perfectly to **Deep NLP** (deep learning). Modern AI doesn't just read word-by-word; it analyzes the whole context of a sentence all at once to get the bigger picture. Because the alien language is non-linear, the team's software has to look at the entire circular symbol simultaneously to understand the true context, which is exactly how deep learning models handle language today.

Comparison

Out of all three, **Deep NLP** is definitely the approach that would help the most in Arrival.

- Rule-based NLP fails pretty quickly on its own because human linguists would have to manually figure out and code every single rule for a completely alien language, which is basically impossible when there's no shared grammar or history. Statistical NLP helps bridge that gap by tracking symbol frequency but it still struggles to grasp the bigger picture when symbols change meaning based on context.

- Deep NLP takes the crown because the heptapods don't express thoughts linearly. They blast out a full circular sentence all at once. By processing the visual symbol as one whole connected image rather than word by word, Deep NLP can handle non-linear grammar, pick up on tiny visual nuances, and figure out the actual context behind complex concepts like realizing the aliens meant "tool" instead of "weapon", which completely changed the trajectory of how the aftermath would have been.

Sources used: *Arrival* via Pluto TV

Samantha — Real-World NLP Tools

- **One communication problem** in *Arrival* is that humans and the aliens do not understand each other's language, which can lead to dangerous misunderstandings.
- **Machine translation** could help by comparing alien symbols and trying to connect them to known meanings. A real-world example is Google Translate.
- **Pattern recognition** could help identify repeated symbols, shapes, or patterns in the aliens' written language. This could make it easier to see which symbols may have similar meanings.
- **Language models** could help analyze previous messages and use context to suggest possible meanings. ChatGPT is a real-world example of a language model.

Section Summary:

These technologies could still misunderstand words or symbols because meaning can depend on context, culture, and intention. In the movie, there is confusion about whether a word means "weapon" or "tool." This is why human experts are still important.

Sources used: *Arrival* (2016), viewed on Pluto TV; Google

Elijah — Final Reflection and Report Assembly

Your section:

One major idea from *Arrival* is how the word "**weapon**" causes fear because humans are not sure what aliens really mean. Louise realizes that language depends on context, not just the direct meaning of a word. The movie shows that culture, cooperation, and understanding matter because people can misunderstand each other even when they are trying to communicate.

Another thing *Arrival* shows is that communication takes patience. Louise does not rush to judge the aliens. She keeps asking questions, studies their symbols, and tries to understand how their language works before making conclusions. That connects to AI because language systems also

need enough information and context before they can give a reliable meaning. A good lesson from the movie is that both people and AI can make mistakes when they assume too much too quickly. The movie also connects to AI because AI can help find patterns and translate language, but it can still miss context or meaning. Just like in *Arrival*, a translation can be technically correct but still leads to the wrong understanding. The main lesson is that people who build or use AI should not depend only on the first answer. Context, culture, and human judgment are still important.

Sources used:

Arrival (2016) - plutotv

IBM – Natural Language Processing (NLP)

Final References

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Villeneuve, D. (Director). (2016). *Arrival* [Film]. Paramount Pictures. Viewed via Pluto TV and YouTube.